

Abstract

We present a biologically motivated, training-free preprocessing pipeline that maps raw images to a 27x compressed discrete edge representation. Motivated by retinal ganglion cells and V1 simple cells, the pipeline chains luminance weighting, Sobel edge detection, sigmoid-based edge probability estimation, and a novel hardcoded 9-template convolution to produce a symbolic edge map in which each 3×3 patch is assigned one of 9 oriented edge types. We demonstrate via human perceptual evaluation and SSIM against Sobel edge probability maps that the representation preserves perceptually relevant structure across diverse image types. Notably, standard pixel-level metrics such as SSIM against raw pixels fail to capture reconstruction quality for discrete compressed representations, a finding with broader implications for evaluation in discrete and tokenised vision systems. We propose this representation as a structured input for downstream models, hypothesising that hardcoding known geometric structure reduces the representational burden and may improve sample efficiency

Introduction

Raw pixels, even after basic pre processing like normalisation, luminance weighing, and more, serve as a poor input representation. # citation They encode no spatial continuity or interconnection between pixels. Biological systems generalise with remarkably few examples, a capability attributed to structured geometric representations built into the early visual pathway and ability to interact with objects both of which together lead to object “grounding”. # citation Contemporary visual models, however, require orders of magnitude more training data to achieve a comparable accuracy, and despite that struggle to encode the semantic meaning of objects, viewpoint changes and generalisation across scale. This paper addresses the first aspect of this biological capability: structure.

The human brain works by first extracting edge information via the network of the retinal ganglion cells and the hierarchical ventral stream, then V1 simple cells fire selectively in response to oriented edges, as demonstrated by Hubel and Wiesel (1959). Higher level features such as colour and texture are integrated in later stages of the ventral stream. The robustness of biological edge detection arises from graded, overlapping tuning curves rather than binary decisions, a principle we replicate through probabilistic edge estimation via a parametrised sigmoid activation prior to discrete encoding. Interestingly, the brain has a centre surround receptive field bias, i.e. spatial resolution is highest in the central visual field and degrades toward the periphery, whereas our method labels edges uniformly across the image.

Classical methods like Sobel and Canny produce continuous gradient maps without symbolic labelling, and while CNNs rediscover Gabor-like structure in early layers, this emerges only with large amounts of training data. We argue for making this structure explicit (A full comparison follows in related work).

Related work:

Sobel (1968): Differs in method and motivation, essentially applies a filter to get continuous edges, intended as a 3×3 Isotropic Gradient Operator for Image Processing, whereas our representation is a discrete map, intended to serve as input for ML models.

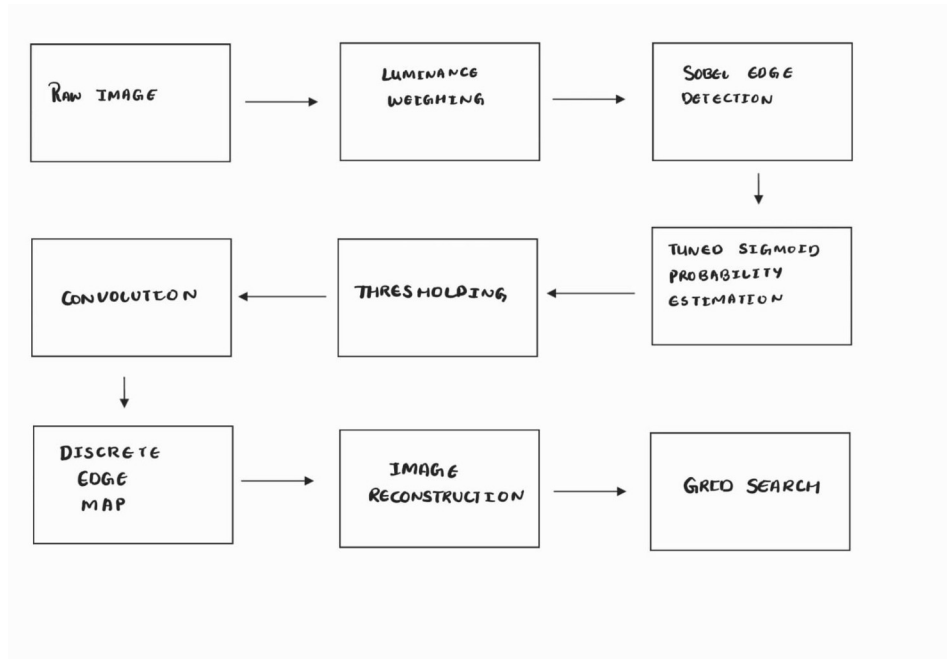
HOG (2005): Again, produces a constant histogram at each pixel and was intended for classical ML, not as a structured input to deep learning models.

Gabor filters: These approximate V1 simple cell responses through bandpass filtering at multiple orientations and scales, but produce continuous multi-channel outputs with no compression. Our method takes the same biological inspiration and reduces it to a minimal discrete basis of 9 symbols, producing a compressed symbolic representation instead.

Hubel and Wiesel (1959): The biological inspiration for V1 cells.

Methodology:

We thought the best way to explain the methodology is walking the reader through the code pipeline.



1. Luminance weighing:

Colours are given unequal importance going into the pipeline, following the brain. (sorry blue).

*Find citation. This is done with a basic multiplication, multiplying each colour channel with its respective weight.

2. Sobel edge detection:

A basic sobel filter is applied to the image. This makes our subsequent steps much more convenient as getting distinct lines through each 3×3 pixel grid brings down the possible edge combinations down from the theoretical maximum of 512 to merely 9 (A 3×3 binary grid has $2^9 = 512$ possible patterns). This reasoning may not be crystal at this point in the pipeline, but will be appreciated better at the next step. #citation

3. Tuned sigmoid probability estimation:

For each pixel, an edge probability is determined using the Sobel filtered image. Why not use raw Sobel values for the next step? That's because our probability is calculated using a sigmoid with two hyper parameters, alpha (controls the slope of the sigmoid) and r (shifts the curve to the right), which gives us significantly more control.

$$f(x) = \left(\frac{1}{1 + e^{-a(x-r)}} \right)$$

Higher values of alpha make small differences in the gradient more important. For example, a gradient of 3.75 gives approximately a probability of 0.076 but a gradient of 4 gives a probability of 0.5 at alpha = 10 and r = 4. (tiny difference in gradient, huge one in probability, yes these are actual values from Desmos). "r" makes it so that higher gradient values are needed for the same probability, but unlike alpha, it is constant for all gradient values.

4. Thresholding:

We introduce a threshold to force the model to decide whether there is an edge or not. We run the simple operation pseudocoded as, if probability of edge > threshold, there is an edge. Else, no edge.

This again, makes our subsequent step much more convenient.

5. Convolution and Discrete edge map:

We hardcode a 3d array mapping 9 different edge possibilities in a three pixel grid (In the code its a 4d tensor to get shapes to cooperate but it's way easier to visualise it as a 3d array). Diagonals (left to right and right to left), columns (left, middle, right), rows (top, middle, bottom) and no edge. This is where all our initial sobel filtering and binarisation works its charm because there's now only about 9 possibilities as to how an edge will run through a 3×3 pixel grid. Without all of that, there would've been 512 edge possibilities, corners, random pixel arrangements and some coherent edges. (Dunno bout you, but coding a 512 element 3d array sounds like something I would rather not do)

We then run a convolution with 9 filters, each discrete edge possibility serving as a 3×3 filter. Since stride is also 3, each non-overlapping 3×3 patch of the image is assigned a single edge label. This reduces both spatial dimensions by a factor of 3, giving a 9x reduction in total elements. Combined with the reduction from 3 colour channels to a single greyscale representation, the total compression relative to the original image is $3 \times 3 \times 3 = 27x$.

Here's one of our filters for easier visualisation:

```
[1, 0, 0],  
[0, 1, 0],  
[0, 0, 1]
```

(The left to right diagonal)

Finally, we find the best match for each 3×3 pixel grid of the image, creating an edge map 9x smaller than our current image in spatial dimensions. As we have also given up our colour information, it is 27x smaller than the original image.

So we now have a labelled edge, one of 9 possible ones at each 3×3 grid in the image.

6. Reconstruction:

We reconstruct the image very simply. Look at each index of our edge map, go to that index (that edge) in our mapping array, and just put it there. This gets us our original dimensions back (minus the colour channels) and plot a reconstructed image.

Interestingly, on our first few runs, we got streaks of diagonal lines across our images. On further inspection, we realised it was because of the fact that wherever there were no images, all convolutions returned 0, and the model ended up choosing practically randomly. We fixed this by adding a small bias to the no edge filter. Arguably, this again is a hyper parameter, but we did not tune it because it seemed to work just fine at a very minimal value.

7. Grid search:

We now must find the right value for the three hyper parameters above, alpha, r, and threshold (one may also consider the luminance weights a hyper parameter but they've already been thoroughly optimised by other people).

We did this by conducting an informal perceptual evaluation in two stages. First, during parameter selection, three observers evaluated reconstructions across parameter combinations to guide grid search decisions. (We had people stare and go hmm... yeah that looks better) Following parameter

selection, ten observers were shown the best reconstruction alongside the original and asked whether the subject was identifiable. All ten confirmed the subject was recognisable. Second, we determined which reconstructions had the highest SSIM score. #find citation. Very satisfyingly, The SSIM scores coincided with our human perceptual study (eyeballing). Interestingly, according to all three of the humans, the top three contenders were all images with $r = 0.4$ and threshold = 0.5 and all values of alpha as 5, 10 and 20. The humans had a hard time deciding which was better and although all concluded it was alpha = 20, it was very close. Whereas, the SSIM scores for the same three reconstructions were significantly different with the score at alpha=5 being much lower than that at alpha=20. This suggests SSIM is more sensitive to sigmoid sharpness than human perception, which may reflect the metric's pixel-level sensitivity to edge softness. Code and results are available in the project repository

Future work

We plan to further imitate biology by digitally mimicking pupil sensitivity so that the model works for dimly lit images, by computing average luminance and adaptively scaling α , r and threshold so that the pipeline can increase edge sensitivity in low-light conditions. We shall also add, the notion of object separation and continuity by incorporating Gestalt-motivated edge continuity detection, where compatible adjacent edge tokens are linked to form coherent object boundaries. And finally we plan to use this representation as tokens to a transformer to determine if it actually reduces training data required for classification tasks and improves efficiency.

Since our model is learning free and all operations are GPU parallelised, it requires very less compute.

Results

We evaluated the images' reconstruction quality and structure preservation on two metrics. One, human perception (arguably the best vision model in existence) and the SSIM metric.

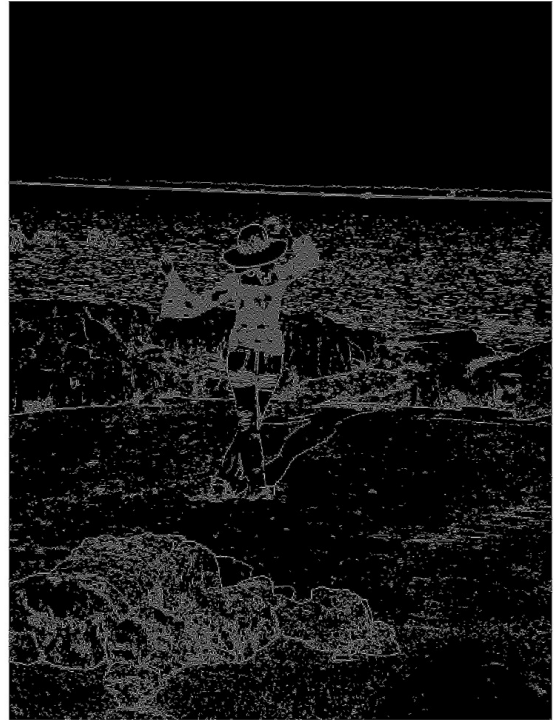
1. Human perception

Across 3 images and 10 people (3 for parameter selection and 10 for confirmation), all agreed that the reconstruction was the same image as the original. The reader may draw their own conclusion from the image below:

Original



Discrete Edge Reconstruction



Despite the challenges of harsh natural lighting, complex structures and smooth transitions across colours, the reconstruction on the right is definitely the same image. (fight me)

2. SSIM

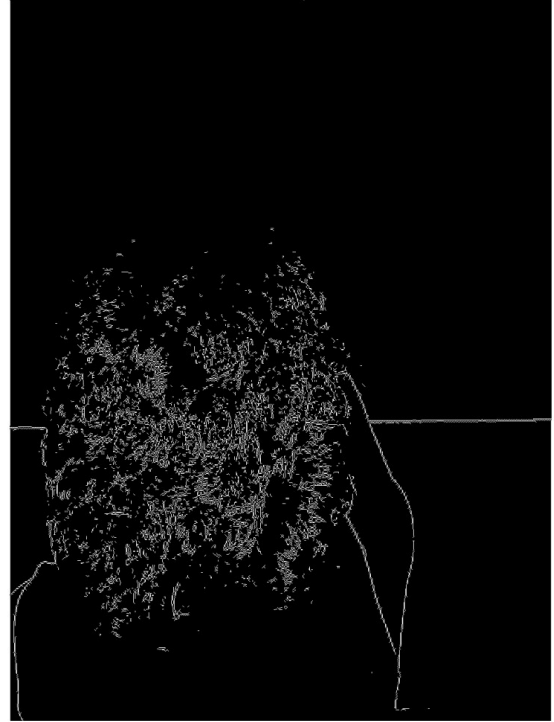
SSIM measures structural similarity between two images. Since we're intentionally discarding information about colour and texture, running SSIM on raw pixel values would be an inaccurate test of our reconstructions. In lieu of that, we ran the SSIM metric on the images' sobel gradient values and our discrete reconstruction. This compares how our discrete 9 edge reconstruction done by labels fares against the continuous edge structure of the original image.

Across three images, here's the results:

Original



Discrete Edge Reconstruction



The red hair got the highest SSIM score, 0.8550 and we believe the visualisation supports the metric.

Original



Discrete Edge Reconstruction



The above image of the dog (noir, who was very uncooperative getting photographed) got the lowest score of 0.4466. That again, is pretty faithful to the quality of the reconstruction.

The beach image got a score of 0.6225

All results reported at $\alpha=20$, $r=0.4$, threshold=0.4, selected via grid search over 48 parameter combinations per image.

Limitations:

The limitations are sufficiently visible in the reconstructions and are also demonstrated in the SSIM scores. The discrete representation struggles with areas with lots of texture (like noir's fur) and also fails to detect any edge whatsoever in dimly lit areas (like the trees in the background of my hair). A proposed fix for these is making the hyper parameters α , r and the threshold sensitive to average luminance (like a digital pupil), but that is beyond the scope of this paper's findings and is left to future work.